1. What is the open-loop voltage gain of an ideal operational amplifier?
(A) Infinite
(B) Zero
(C) 1000
(D) 1.0

2. What is the input impedance of an ideal operational amplifier?
(A) Infinite
(B) Zero
(C) 50 Ohms
(D) 10 kOhms

3. What is the output impedance of an ideal operational amplifier?
(A) Zero
(B) Infinite
(C) 100 Ohms
(D) 1 kOhm

4. What is the bandwidth of an ideal operational amplifier?
(A) Infinite
(B) 1 MHz
(C) 10 kHz
(D) Zero

5. The Common Mode Rejection Ratio (CMRR) of an ideal op-amp is:
(A) Infinite
(B) Zero
(C) 100 dB
(D) 1.0

6. Slew rate of an ideal operational amplifier is:
(A) Infinite
(B) 1 V/microsecond
(C) Zero
(D) 10 V/ms

7. Slew rate is physically measured in units of:
(A) Volts per microsecond (V/us)
(B) Volts per second
(C) Amperes per second
(D) Watts

8. The pin number for the inverting input of standard 8-pin 741 Op-Amp IC is:
(A) Pin 2
(B) Pin 3
(C) Pin 6
(D) Pin 7

9. The pin number for the non-inverting input of 741 Op-Amp IC is:
(A) Pin 3
(B) Pin 2
(C) Pin 6
(D) Pin 4

10. The output pin of the 741 Op-Amp IC is:
(A) Pin 6
(B) Pin 2
(C) Pin 3
(D) Pin 7

11. The concept of 'Virtual Ground' at the inverting terminal is valid only when:
(A) Negative feedback is applied and op-amp is in linear range
(B) Positive feedback is applied
(C) Op-amp is in saturation
(D) Supply is disconnected

12. The closed-loop voltage gain Av of an inverting amplifier with input resistor R1 and feedback Rf is:
(A) - Rf / R1
(B) 1 + Rf / R1
(C) Rf / R1
(D) - R1 / Rf

13. The closed-loop voltage gain Av of a non-inverting amplifier with input R1 and feedback Rf is:
(A) 1 + (Rf / R1)
(B) - Rf / R1
(C) Rf / R1
(D) 1 - (Rf / R1)

14. A Voltage Follower (Unity Gain Buffer) has a closed-loop voltage gain of:
(A) 1.0 (Unity)
(B) Zero
(C) -1.0
(D) Infinite

15. The primary application of an op-amp voltage follower is:

(A) Impedance matching and buffer isolation (high Rin, zero Rout)
(B) Voltage amplification
(C) Phase inversion
(D) Filtering

16. In an inverting summing amplifier, the output voltage is the:

(A) Negative weighted sum of the input voltages
(B) Product of inputs
(C) Positive sum of inputs
(D) Difference of inputs

17. An ideal difference amplifier amplifies the:

(A) Difference between two input voltages while rejecting common-mode signals
(B) Sum of inputs
(C) Average of inputs
(D) High frequencies only

18. An op-amp basic integrator replaces the feedback resistor Rf with a:

(A) Capacitor C
(B) Inductor L
(C) Diode
(D) Zener diode

19. An op-amp basic differentiator places a capacitor C in:

(A) The input branch in series with the inverting terminal
(B) The feedback loop
(C) Parallel with load
(D) The power supply

20. The output voltage of an ideal op-amp integrator is proportional to the:

(A) Time integral of input voltage: - (1 / RC) * Integral(Vin dt)
(B) Derivative of input
(C) Square of input
(D) Logarithm of input

21. The output voltage of an ideal op-amp differentiator is proportional to the:

(A) Time derivative of input voltage: - R * C * (dVin / dt)
(B) Integral of input
(C) Exponential of input
(D) Constant

22. A basic op-amp differentiator is prone to instability and noise because its gain:

(A) Increases linearly with frequency (Av = -omega * R * C)
(B) Decreases with frequency
(C) Is constant
(D) Is zero at high frequency

23. To stabilize a practical differentiator at high frequencies, one must add a:

(A) Resistor in series with input capacitor and capacitor in parallel with Rf
(B) Diode in series
(C) Zener diode
(D) Inductor

24. A comparator compares an input voltage with a reference voltage and outputs:

(A) +Vsat or -Vsat (binary high or low)
(B) A linear amplified voltage
(C) Zero always
(D) A pure sine wave

25. A Zero-Crossing Detector is a comparator where the reference voltage Vref is:

(A) 0 V (Ground)
(B) +5 V
(C) -5 V
(D) +Vcc

26. A Schmitt Trigger employs which type of feedback to establish hysteresis?

(A) Positive Feedback
(B) Negative Feedback
(C) No feedback
(D) Current feedback only

27. The purpose of hysteresis in a Schmitt trigger is to:

(A) Prevent erratic false switching caused by input noise
(B) Increase gain
(C) Reduce power consumption
(D) Invert phase

28. An active filter contains active elements (op-amps) along with:

(A) Resistors and capacitors (no inductors required)
(B) Bulky iron chokes
(C) Permanent magnets
(D) Mechanical springs

29. The roll-off rate of a First-Order active low-pass filter above its cutoff frequency is:

(A) -20 dB / decade (-6 dB / octave)
(B) -40 dB / decade
(C) -60 dB / decade
(D) -10 dB / decade

30. The roll-off rate of a Second-Order active low-pass filter is:

(A) -40 dB / decade (-12 dB / octave)
(B) -20 dB / decade
(C) -80 dB / decade
(D) Zero

31. A Butterworth filter characteristic is famous for having a:

(A) Maximally flat passband response (zero ripple)
(B) Equiripple passband
(C) Steepest roll-off transition
(D) Linear phase only

32. A Chebyshev filter provides a steeper roll-off than Butterworth at the expense of:

(A) Ripple in the passband
(B) Lower gain
(C) Infinite noise
(D) Instability

33. A Band-Pass Filter (BPF) allows frequencies to pass:

(A) Within a specified band between lower cutoff fL and upper cutoff fH
(B) Above cutoff only
(C) Below cutoff only
(D) All frequencies

34. A Band-Reject (Notch) Filter is specifically designed to eliminate:

(A) A single specific interfering frequency (such as 50 Hz power-line hum)
(B) All audio frequencies
(C) DC voltages
(D) High frequencies

35. An All-Pass Filter maintains constant magnitude across all frequencies while:

(A) Shifting phase systematically from 0 to -180 degrees
(B) Blocking DC
(C) Doubling amplitude
(D) Inverting polarity only

36. An Instrumentation Amplifier typically consists of how many op-amps?

(A) 3 Op-Amps
(B) 1 Op-Amp
(C) 2 Op-Amps
(D) 4 Op-Amps

37. The primary advantage of an Instrumentation Amplifier is:

(A) Extremely high CMRR and very high input impedance for differential sensors
(B) Low cost
(C) Operates without DC supply
(D) High distortion

38. The gain of a 3-Op-Amp Instrumentation Amplifier is conveniently adjusted by changing:

(A) A single external gain resistor (Rg)
(B) All seven resistors simultaneously
(C) Supply voltage
(D) Input capacitance

39. An Astable Multivibrator constructed with an op-amp generates a continuous:

(A) Square wave
(B) Sine wave
(C) Triangular wave
(D) Sawtooth wave

40. A Monostable Multivibrator produces:

(A) A single output pulse of fixed duration when triggered
(B) Continuous square wave
(C) Two stable states
(D) Zero output

41. The Gain-Bandwidth Product (GBWP) of an internally compensated op-amp is:

(A) A constant value across all closed-loop gain settings
(B) Proportional to gain
(C) Inversely proportional to bandwidth squared
(D) Zero

42. If an op-amp has GBWP = 1 MHz, its closed-loop 3-dB bandwidth with a gain of 100 is:

(A) 10 kHz
(B) 100 kHz
(C) 1 MHz
(D) 100 Hz

43. Input Offset Voltage Vio of an op-amp is defined as the differential input voltage required to:

(A) Force the output voltage to exactly zero
(B) Saturate the output
(C) Trigger the Schmitt switch
(D) Maximize gain

44. Input Bias Current Ib is defined as the:

(A) Average of the two DC input terminal currents: (Ib1 + Ib2) / 2
(B) Difference between inputs
(C) Current from supply pins
(D) Maximum output current

45. Input Offset Current Iio is defined as the:

(A) Absolute difference between the two input bias currents: |Ib1 - Ib2|
(B) Sum of bias currents
(C) Zero
(D) Current at inverting pin

46. To minimize output DC voltage offset caused by input bias currents in an inverting amplifier, R_comp at non-inverting terminal should equal:

(A) R1 || Rf = (R1 * Rf) / (R1 + Rf)
(B) R1 + Rf
(C) Rf - R1
(D) Zero

47. Common Mode Rejection Ratio in decibels is calculated by:

(A) 20 * log10(|Ad| / |Acm|)
(B) 10 * log10(Ad / Acm)
(C) Ad / Acm
(D) 20 * log10(Acm / Ad)

48. If differential gain Ad = 10^5 and common-mode gain Acm = 10, the CMRR in dB is:

(A) 80 dB
(B) 100 dB
(C) 40 dB
(D) 10,000 dB

49. Full-power bandwidth (f_max) of an op-amp delivering peak output voltage Vp without slew-rate distortion is:

(A) SR / (2 * pi * Vp)
(B) 2 * pi * SR * Vp
(C) Vp / SR
(D) SR / Vp

50. An op-amp with slew rate 0.5 V/us supplies a peak sinusoidal output of 10 V. Maximum undistorted frequency is:

(A) 7.96 kHz
(B) 50 kHz
(C) 100 kHz
(D) 1.59 kHz

51. In a non-inverting Schmitt trigger with feedback resistor Rf and input resistor R1, threshold voltages are:

(A) +- Vsat * (R1 / Rf)
(B) +- Vsat * (Rf / R1)
(C) +- Vsat * (1 + Rf/R1)
(D) Zero

52. Hysteresis width (VH) of the above Schmitt trigger is:

(A) 2 * Vsat * (R1 / Rf)
(B) Vsat * (R1 / Rf)
(C) Vsat
(D) Zero

53. A Precision Half-Wave Rectifier (Super Diode) eliminates the 0.7 V diode barrier potential by placing the diode:

(A) Inside the negative feedback loop of the op-amp
(B) In series with power supply
(C) Across inputs
(D) In parallel with ground

54. In a basic inverting op-amp integrator with input R and feedback C, cutoff frequency where gain is unity (0 dB) is:

(A) f_unity = 1 / (2 * pi * R * C)
(B) 2 * pi * R * C
(C) R * C
(D) 1 / (R * C)

55. A practical op-amp integrator places a large shunt resistor Rf in parallel with C to:

(A) Limit DC open-loop gain and prevent drift into saturation
(B) Increase high frequency gain
(C) Invert output
(D) Provide positive feedback

56. A Sallen-Key second-order low-pass active filter topology uses:

(A) Two resistors, two capacitors, and one op-amp in non-inverting configuration
(B) Three op-amps
(C) Inductors and capacitors
(D) Transformers

57. The cutoff frequency fc of an equal-component Sallen-Key low-pass filter (R1=R2=R, C1=C2=C) is:

(A) 1 / (2 * pi * R * C)
(B) 1 / (4 * pi * R * C)
(C) 2 * pi * R * C
(D) R / C

58. To achieve maximally flat Butterworth response (damping ratio zeta = 0.707, Q = 0.707) in a Sallen-Key low-pass filter, closed-loop gain K must be:

(A) K = 3 - sqrt(2) approx 1.586
(B) 1.0
(C) 2.0
(D) 3.0

59. If closed-loop gain K of a Sallen-Key low-pass filter is increased to 3.0 or greater, the circuit:

(A) Becomes unstable and breaks into self-sustained oscillation
(B) Attains infinite bandwidth
(C) Becomes an ideal filter
(D) Burns out

60. A State-Variable (KHN) Active Filter simultaneously provides outputs for:

(A) Low-Pass, High-Pass, and Band-Pass responses
(B) Low-Pass only
(C) All-Pass only
(D) Rectification

61. The Wein-Bridge Oscillator utilizes an op-amp with both positive and negative feedback. Oscillation frequency is:

(A) f0 = 1 / (2 * pi * R * C)
(B) f0 = 1 / [2 * pi * sqrt(6) * R * C]
(C) f0 = 2 * pi * R * C
(D) f0 = 1 / (R * C)

62. In a Wein-Bridge oscillator, the closed-loop non-inverting gain required to sustain oscillations (Barkhausen criterion) is:

(A) Exactly 3.0 (1 + Rf/R1 = 3 => Rf = 2 * R1)
(B) 1.0
(C) 29.0
(D) 10.0

63. An RC Phase-Shift Oscillator using an inverting op-amp requires a 3-stage RC ladder network producing phase shift of:

(A) 180 degrees (60 degrees per stage)
(B) 90 degrees
(C) 360 degrees
(D) 0 degrees

64. The minimum inverting amplifier gain required to maintain sustained oscillations in an RC phase-shift oscillator is:

(A) 29.0 (Rf / R1 = 29)
(B) 3.0
(C) 1.0
(D) 10.0

65. The oscillation frequency of an RC Phase-Shift Oscillator with identical R and C stages is:

(A) f0 = 1 / [2 * pi * sqrt(6) * R * C]
(B) 1 / (2 * pi * R * C)
(C) sqrt(6) / (2 * pi * R * C)
(D) 1 / (R * C)

66. In an op-amp logarithmic amplifier, the feedback element connected across the inverting terminal is a:

(A) BJT transistor (or forward-biased diode)
(B) Resistor
(C) Capacitor
(D) Zener diode

67. The output voltage of an op-amp log amplifier is proportional to:

(A) - Vt * ln(Vin / (R * Is))
(B) exp(Vin)
(C) Vin^2
(D) sqrt(Vin)

68. In an anti-log (exponential) amplifier, the non-linear BJT element is placed in the:

(A) Input branch
(B) Feedback branch
(C) Supply pin
(D) Output terminal

69. An analog multiplier can be constructed by combining:

(A) Two Log amplifiers, a Summing amplifier, and an Anti-Log amplifier
(B) Two Integrators
(C) Three Schmitt triggers
(D) A Wein bridge

70. Peak detector circuits capture and hold the peak value of an analog input voltage using an:

(A) Op-amp, diode, and low-leakage storage capacitor
(B) Inductor
(C) Comparator only
(D) Zener diode

71. A Sample-and-Hold (S/H) circuit uses an analog FET switch and storage capacitor to:

(A) Freeze an analog voltage level constant during ADC conversion
(B) Filter noise
(C) Amplify AC
(D) Generate pulses

72. A Current-to-Voltage Converter (Transimpedance Amplifier) converts input current Iin into output voltage Vout = :

(A) - Iin * Rf
(B) - Iin / Rf
(C) Iin * R1
(D) Zero

73. Transimpedance amplifiers are universally used in front-end circuits for:

(A) Photodiode light detection and optical receivers
(B) Power supplies
(C) Audio speakers
(D) Motor controllers

74. In a 3-Op-Amp Instrumentation Amplifier, the input buffer stage provides:

(A) High differential gain and unity common-mode gain, vastly boosting CMRR
(B) Zero gain
(C) Pure attenuation
(D) Phase inversion

75. The differential gain of a 3-Op-Amp Instrumentation Amplifier with identical input stage resistors R and gain resistor Rg is:

(A) Av = [1 + (2 * R / Rg)] * (R3 / R2)
(B) 1 + R/Rg
(C) 2 * R / Rg
(D) R3 / R2

76. Internal frequency compensation in the 741 op-amp is achieved by a 30 pF capacitor creating:

(A) Pole splitting via Miller effect, establishing a dominant low-frequency pole at ~5 Hz
(B) Zero cancellation
(C) High pass response
(D) Feedback cancellation

77. The open-loop transfer function of an internally compensated op-amp is modeled as:

(A) A(s) = A0 / (1 + s / omega_p1)
(B) A(s) = A0 * s
(C) A(s) = 1 / s^2
(D) A(s) = A0

78. An uncompensated op-amp has multiple high-frequency poles. Connecting heavy resistive feedback can cause:

(A) Instability and spontaneous oscillation due to excessive phase lag (> 180 deg) at unity gain crossover
(B) Instant burnout
(C) Zero gain
(D) Pure noise

79. Phase Margin (PM) of a feedback amplifier is defined as:

(A) 180 deg + Phase of Loop Gain T(j*omega) at the unity-gain frequency omega_unity
(B) 180 deg - Phase
(C) Phase at DC
(D) 90 deg

80. For acceptable transient settling with minimal overshoot (< 5%), the phase margin should be at least:

(A) 45 to 60 degrees
(B) 0 degrees
(C) 10 degrees
(D) 180 degrees

81. In a Tow-Thomas Biquad active filter, the circuit consists of:

(A) Two integrators and one inverter connected in a closed feedback loop
(B) Three differentiators
(C) Four comparators
(D) Two adders

82. Switched-Capacitor Filters simulate high-precision resistors using MOS switches and capacitors: R_eq = :

(A) 1 / (f_clk * C)
(B) f_clk * C
(C) f_clk / C
(D) 1 / (f_clk^2 * C)

83. The transfer function of a Normalized Second-Order Butterworth Low-Pass Filter is:

(A) H(s) = 1 / (s^2 + sqrt(2)*s + 1)
(B) H(s) = 1 / (s^2 + s + 1)
(C) H(s) = 1 / (s^2 + 2*s + 1)
(D) H(s) = 1 / (s + 1)

84. In a second-order active filter, the pole quality factor Q is related to damping ratio zeta by:

(A) Q = 1 / (2 * zeta)
(B) Q = 2 * zeta
(C) Q = zeta^2
(D) Q = 1 / zeta

85. For a maximally flat Butterworth response, the pole Quality factor Q is:

(A) 1 / sqrt(2) approx 0.7071
(B) 1.0
(C) 0.5
(D) 2.0

86. In a Bessel active filter, the primary optimized characteristic is:

(A) Maximally linear phase response (maximally flat group delay, zero waveform dispersion)
(B) Steepest roll-off
(C) Minimum capacitor size
(D) Zero DC gain

87. An op-amp Precision Full-Wave Rectifier (Absolute Value Circuit) can be constructed using:

(A) Two op-amps, two signal diodes, and precision matched resistors
(B) One diode and one inductor
(C) A transformer
(D) A Schmitt trigger

88. In a precision clamp circuit (DC restorer), the output waveform:

(A) Restores the DC baseline of an AC signal to a fixed reference potential
(B) Clips the signal symmetrically
(C) Rectifies to DC
(D) Inverts phase

89. A Triangular Wave Generator is constructed by cascading:

(A) An Op-Amp Schmitt Trigger and an Op-Amp Integrator in a closed loop
(B) Two differentiators
(C) A Wein bridge and filter
(D) Two comparators

90. In the Triangular-Square Wave generator above, the oscillation frequency is:

(A) f0 = R2 / [4 * R1 * R * C]
(B) 1 / (2 * pi * R * C)
(C) R / C
(D) 1 / [4 * R * C]

91. The Voltage-to-Frequency Converter (VFC) generates an output pulse train whose frequency is:

(A) Linearly proportional to the applied DC input voltage
(B) Inversely proportional to voltage
(C) Exponential
(D) Zero

92. An op-amp Gyrator circuit (active inductor simulator) synthesizes equivalent grounded inductance L = :

(A) R1 * R3 * R5 * C / (R2 * R4)
(B) R^2 * C
(C) C / R
(D) R * C^2

93. Frequency-Dependent Negative Resistor (FDNR) or D-element in active filter synthesis has impedance:

(A) Z(s) = 1 / (s^2 * D)
(B) Z(s) = s^2 * D
(C) Z(s) = - R
(D) Z(s) = s * D

94. Using Bruton's 1/s Impedance Scaling Transformation, a passive R-L-C low-pass ladder transforms into:

(A) Resistors -> Capacitors, Inductors -> Resistors, Capacitors -> FDNRs
(B) All inductors
(C) All capacitors
(D) All resistors

95. The main advantage of transforming an RLC ladder into an active FDNR filter is that:

(A) It avoids all floating inductors, requiring only grounded active FDNR elements
(B) It reduces power to zero
(C) It eliminates op-amps
(D) It operates on DC only

96. Chopper-Stabilized (Zero-Drift) Op-Amps eliminate DC offset voltage and 1/f flicker noise by:

(A) Modulating the input signal to AC, amplifying, and synchronously demodulating back to DC
(B) Using liquid nitrogen cooling
(C) Eliminating feedback
(D) Adding giant capacitors

97. Current-Feedback Operational Amplifiers (CFOAs) differ fundamentally from Voltage-Feedback Op-Amps because:

(A) Their closed-loop bandwidth is largely independent of closed-loop gain
(B) Gain is strictly 1
(C) Input impedance is infinite
(D) They cannot amplify AC

98. The inverting input of a Current-Feedback Op-Amp (CFOA) is internally connected to:

(A) The output of an internal unity-gain buffer from the non-inverting input (low impedance terminal)
(B) A high-Z gate
(C) Ground
(D) Power supply

99. The slew rate of a Current-Feedback Op-Amp is typically:

- **(A)** Extremely high (> 1000 V/microsecond) because charging current is on-demand and not limited by a tail current source
- **(B)** Low (~0.1 V/us)
- **(C)** Zero
- **(D)** 10 V/us

100. An active Notch Filter constructed with a Twin-T network has notch frequency f0 equal to:

- **(A)** f0 = 1 / (2 * pi * R * C)
- **(B)** f0 = 1 / (4 * pi * R * C)
- **(C)** f0 = sqrt(2) / (pi * R * C)
- **(D)** f0 = 2 * pi / (R * C)

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Derive the closed-loop voltage gain expressions for Inverting and Non-Inverting operational amplifier configurations using the Virtual Ground concept.

Q2. Analyze the 3-Op-Amp Instrumentation Amplifier: Derive the closed-loop differential gain and explain its superior Common Mode Rejection Ratio.

Q3. Analyze the basic Op-Amp Integrator and Differentiator, explain their frequency-domain limitations, and design practical compensated versions.

Q4. Analyze the Inverting and Non-Inverting Op-Amp Schmitt Trigger: Derive Upper and Lower Threshold Voltages and Hysteresis width.

Q5. Analyze Precision Rectifiers (Half-Wave and Full-Wave Absolute Value Circuits), explaining how diode forward drop is eliminated.

Q6. Derive the transfer function and design equations of a Second-Order Low-Pass Sallen-Key Active Butterworth Filter.

Q7. Explain the KHN (Kerwin-Huelsman-Newcomb) State-Variable Active Filter and its biquadratic tuning properties.

Q8. Analyze the Wein-Bridge Oscillator: Derive the condition for oscillation and frequency of oscillation using Barkhausen Criterion.

Q9. Explain the Op-Amp Astable Multivibrator (Square Wave Generator), deriving the formula for oscillation frequency and duty cycle.

Q10. Explain the Triangular-Square Wave Generator circuit, deriving formulas for peak-to-peak amplitude and frequency.

Q11. Explain Slew Rate and derive the expression for Full-Power Bandwidth (f_max) of an operational amplifier.

Q12. Explain Active Filter Approximation Functions: Compare Butterworth, Chebyshev, Bessel, and Elliptic filters.

Q13. Explain Switched-Capacitor Filters (SCF), deriving the equivalent resistance of a clocked capacitor.

Q14. Explain Logarithmic and Anti-Logarithmic Op-Amp Amplifiers, detailing temperature compensation techniques.

Q15. Explain Analog Multipliers and Dividers constructed using Log, Anti-Log, and Summing operational amplifiers.

Q16. Analyze Op-Amp Frequency Compensation: Explain Dominant Pole Compensation, pole-splitting, and phase margin.

Q17. Explain Current-Feedback Operational Amplifiers (CFOA), comparing architecture, slew rate, and gain-bandwidth independence with VFOA.

Q18. Design a 50 Hz Active Notch Filter using a Twin-T network in an op-amp feedback configuration.

Q19. Explain the Tow-Thomas Biquad active filter and derive its transfer function.

Q20. Explain Bruton's FDNR (Frequency-Dependent Negative Resistor) Transformation for realizing active ladder filters without inductors.

Q21. Explain Chopper-Stabilized (Zero-Drift) Operational Amplifiers and their noise reduction mechanisms.

Q22. Explain Sample-and-Hold (S/H) Amplifiers, defining acquisition time, aperture jitter, and droop rate.

Q23. Explain the Wien-Bridge Oscillator Amplitude Stabilization mechanisms (Diodes, JFET, and PTC Thermistors).

Q24. Explain Phase Margin and Gain Margin in feedback amplifiers using Bode Plots.

Q25. Analyze the Op-Amp Voltage-to-Current Converter (Transconductance Amplifier) for Floating and Grounded Loads.

Q26. Explain the Phase-Locked Loop (PLL), detailing Phase Detector, Loop Filter, and Voltage-Controlled Oscillator (VCO).

Q27. Analyze the State-Variable Band-Pass Filter, deriving formulas for center frequency, bandwidth, and peak gain.

Q28. Explain the Monostable Multivibrator using an Op-Amp, deriving the output pulse width.

Q29. Explain Active Inductor Simulation using Antoniou's General Impedance Converter (GIC).

Q30. Describe Instrumentation Operational Amplifiers in Biomedical Engineering: Bio-potential acquisition (ECG/EEG) and Right-Leg Drive circuits.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1: [A]
Q2: [A]
Q3: [A]
Q4: [A]
Q5: [A]
Q6: [A]
Q7: [A]
Q8: [A]
Q9: [A]
Q10: [A]
Q11: [A]
Q12: [A]
Q13: [A]
Q14: [A]
Q15: [A]
Q16: [A]
Q17: [A]
Q18: [A]
Q19: [A]
Q20: [A]
Q21: [A]
Q22: [A]
Q23: [A]
Q24: [A]
Q25: [A]
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Q62: [A]
Q63: [A]
Q64: [A]
Q65: [A]
Q66: [A]
Q67: [A]
Q68: [A]
Q69: [A]
Q70: [A]
Q71: [A]
Q72: [A]
Q73: [A]
Q74: [A]
Q75: [A]
Q76: [A]
Q77: [A]
Q78: [A]
Q79: [A]
Q80: [A]
Q81: [A]
Q82: [A]
Q83: [A]
Q84: [A]
Q85: [A]
Q86: [A]
Q87: [A]
Q88: [A]
Q89: [A]
Q90: [A]
Q91: [A]
Q92: [A]
Q93: [A]
Q94: [A]
Q95: [A]
Q96: [A]
Q97: [A]
Q98: [A]
Q99: [A]
Q100: [A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

[illegible]

- Because no current enters op-amp inputs, current I_g flows through both feedback resistors R in series:

Output of A1: Vo1 = V1 + I_g * R = V1 + R * (V1 - V2) / Rg.

Output of A2: Vo2 = V2 - I_g * R = V2 - R * (V1 - V2) / Rg.

- Differential output of input buffer stage:

Vo1 - Vo2 = (V1 - V2) + 2 * R * (V1 - V2) / Rg = (V1 - V2) * [1 + (2 * R / Rg)].

- The buffer stage amplifies differential signal (V1 - V2) by factor [1 + 2*R/Rg].
- For common-mode signals (V1 = V2 = Vcm): I_g = 0; hence Vo1 = Vo2 = Vcm. Common-mode gain of input stage is strictly 1.0!
- Output Difference Stage (A3):

Vout = (R3 / R2) * (Vo2 - Vo1).

- Total Overall Closed-Loop Gain:

Vout = (R3 / R2) * [1 + (2 * R / Rg)] * (V2 - V1).

3. Why Instrumentation Amplifiers Have Superior CMRR:

- The first stage amplifies differential signals with high gain while passing common-mode noise with unity gain (gain = 1).
- Thus, the ratio of differential signal to common-mode signal is multiplied by [1 + 2*R/Rg] BEFORE reaching the difference stage.
- Provides CMRR exceeding 100-120 dB even with minor resistor tolerances; extremely high input impedance prevents sensor loading.

[Detailed Step-by-Step Answer]:
Op-Amp Schmitt Trigger and Hysteresis Analysis:
1. Inverting Schmitt Trigger:
- Input voltage Vin is applied to the INVERTING terminal.
- Positive feedback voltage divider (R1 from non-inverting terminal to ground, R2 from non-inverting terminal to output Vout).
- Output saturates at either +Vsat or -Vsat.
- Threshold Voltages at Non-Inverting Terminal (V+):
By voltage divider: V+ = Vout * [R1 / (R1 + R2)].
- When Vout = +Vsat:
Upper Threshold Voltage: V_UT = + Vsat * [R1 / (R1 + R2)].
- When Vout = -Vsat:
Lower Threshold Voltage: V_LT = - Vsat * [R1 / (R1 + R2)].
- Switching Dynamics:
* Starting with Vin < V_LT: Vout = +Vsat.
* As Vin increases, Vout remains +Vsat until Vin crosses V_UT. Once Vin > V_UT, output snaps instantaneously to -Vsat.
* To switch back, Vin must decrease all the way below V_LT.
- Hysteresis Width:
V_H = V_UT - V_LT = 2 * Vsat * [R1 / (R1 + R2)].
2. Non-Inverting Schmitt Trigger:
- Inverting terminal grounded (0 V).
- Input Vin connected via R1 to non-inverting terminal; positive feedback R2 from Vout to non-inverting terminal.
- Switching occurs when V+ crosses 0 V:
V+ = (Vin * R2 + Vout * R1) / (R1 + R2) = 0 => Vin * R2 = - Vout * R1.
Vin = - Vout * (R1 / R2).
- Upper Threshold (switching from -Vsat to +Vsat): V_UT = + Vsat * (R1 / R2).
- Lower Threshold (switching from +Vsat to -Vsat): V_LT = - Vsat * (R1 / R2).
- Hysteresis: V_H = 2 * Vsat * (R1 / R2).
3. Purpose of Hysteresis:
- Creates a deadband of width V_H. Any noise spikes smaller than V_H superimposed on slow or noisy analog waveforms cannot trigger false output transitions.

>>Solution to Question 5:

[Problem Statement]: Analyze Precision Rectifiers (Half-Wave and Full-Wave Absolute Value Circuits), explaining how diode forward drop is eliminated.

3. Precision Full-Wave Rectifier (Absolute Value Circuit):
- Combines a precision half-wave inverting rectifier with a summing amplifier.
- Output voltage: Vout = |Vin| for all positive and negative input swings.

[illegible]

- By summing the HPF and LPF outputs with a fourth op-amp, a Notch (Band-Reject) or All-Pass filter is obtained.

3. Independent Tuning Capabilities:

- Natural Frequency: omega_0 = 1 / (R * C), tuned by adjusting integrator resistors simultaneously.
- Quality factor Q: Adjusted independently by changing a single damping resistor in the summing amplifier without altering omega_0!
- Extremely low component sensitivity, robust against parasitic stray capacitances.

[illegible]

- Upper Threshold Voltage: V_UT = beta * (+Vsat) = + beta * Vsat.
- Lower Threshold Voltage: V_LT = beta * (-Vsat) = - beta * Vsat,
where feedback fraction beta = R1 / (R1 + R2).

3. Capacitor Charging and Discharging Transient Derivation:

- Suppose Vout = +Vsat. Capacitor C charges toward +Vsat through resistor R from initial voltage -beta*Vsat:
vc(t) = V_target + [V_initial - V_target] * exp(- t / RC)
= +Vsat + [-beta*Vsat - Vsat] * exp(- t / RC)
= Vsat * [1 - (1 + beta) * exp(- t / RC)].
- When vc(t) reaches +beta*Vsat at t = T1, inverting terminal voltage exceeds non-inverting terminal, snapping output to -Vsat.
- Equating:
beta * Vsat = Vsat * [1 - (1 + beta) * exp(- T1 / RC)].
(1 + beta) * exp(- T1 / RC) = 1 - beta.
exp(T1 / RC) = (1 + beta) / (1 - beta) => T1 = R * C * ln[(1 + beta) / (1 - beta)].
- The discharge half-cycle T2 is completely symmetrical: T2 = T1.

4. Total Period and Frequency:

- Total period: T = T1 + T2 = 2 * R * C * ln[(1 + beta) / (1 - beta)].
- Frequency of oscillation:
f_0 = 1 / T = 1 / [2 * R * C * ln((1 + beta) / (1 - beta))].
- Special Case: If R1 = 1.16 * R2, beta = 0.462 and ln[(1+beta)/(1-beta)] = 1.0:
T = 2 * R * C, and f_0 = 1 / (2 * R * C).

>> Solution to Question 10:

[Problem Statement]: Explain the Triangular-Square Wave Generator circuit, deriving formulas for peak-to-peak amplitude and frequency.

>> Solution to Question 12: [illegible] non-linearity.

[illegible] techniques.

- Monolithic matching cancels the volatile Is term completely!
- PTC Resistor Network: The remaining linear temperature dependence of Vt is eliminated by passing the signal through a non-inverting amplifier containing a Positive Temperature Coefficient (PTC) thermistor with temperature coefficient +3300 ppm/deg C, producing an overall temperature-invariant logarithmic output.

>>Solution to Question 16 [illegible] margin.

- Miller compensation ensures the open-loop response rolls off at a stable -20 dB/decade all the way down to unity-gain crossover, maintaining Phase Margin > 60 degrees for unconditional stability.

[illegible] gain-bandwidth independence with V

[illegible] ladder filters without inductors.

- Conventional CMOS op-amps suffer from substantial 1/f (flicker) noise and DC input offset voltage drift (~microvolts/deg C) due to random transistor mismatch.

2. Chopping Principle:

- Modulation: The input DC/low-frequency signal is chopped (multiplied by a square wave carrier at clock frequency f_ch ~100 kHz), modulating the signal up to high frequency f_ch.
- Amplification: The modulated signal is amplified by the op-amp stage. Crucially, the amplifier's internal DC offset and 1/f noise remain unmodulated at DC.
- Demodulation: A synchronous second chopper demodulates the amplified signal back down to baseband (DC).
- Offset and Noise Shift: The internal offset and 1/f noise pass through the second chopper and are modulated UP to f_ch.
- Low-Pass Filtering: A post-filter strips away the chopped high-frequency noise spikes at f_ch.

3. Performance Outcome:

- Achieves near-zero offset voltage (< 1 microvolt) and virtually zero temperature drift (< 0.005 uV/deg C) with complete elimination of 1/f flicker noise.

>> Solution to Question 22:

[Problem Statement] Explain the Wien-Bridge Oscillator Amplitude Stabilization mechanisms (Diodes, JFET and PTC Thermistors).

[Problem Statement] Explain Sample-and-Hold (S/H) Amplifiers, acquisition time and droop rate.

- Tungsten has positive temperature coefficient: resistance R1 rises, decreasing amplifier gain Av = 1 + Rf/R1 back to exactly 3.0.

3. Back-to-Back Diode Shunt:

- Two antiparallel diodes in series with resistor placed in parallel with feedback resistor Rf.
- For small signals, diodes are OFF, maintaining Av > 3.0 to initiate oscillations.
- When amplitude exceeds 0.7 V, diodes conduct, reducing effective feedback resistance Rf and clamping Av at 3.0.

4. JFET Voltage-Controlled Resistor:

- JFET operated in ohmic region replaces R1. A peak detector rectifies output and feeds negative gate-source voltage Vgs to JFET to dynamically modulate channel resistance.

>>Solution to Question 24: Analyze Phase Margin and Gain Margin in feedback amplifiers using Bode Plots.

[Problem Statement: Explain the Op-Amp Voltage-to-Current Converter (Transconductance Amplifier) for Floating and Grounded Loads.

[illegible]

- Diode D1 clamps capacitor C to ~0.7 V, preventing further charging.
- Voltage divider establishes positive threshold V_UT = beta * (+Vsat) > 0.7 V.

2. Triggering and Quasi-Stable State:

- A negative trigger pulse applied to non-inverting terminal drives V+ below 0.7 V.
- Op-amp output snaps instantaneously to -Vsat (Quasi-Stable State).
- Positive feedback establishes negative threshold V_LT = - beta * Vsat.
- Diode D1 becomes reverse biased. Capacitor C begins charging exponentially toward -Vsat through resistor R.

3. Pulse Width Derivation:

- Capacitor voltage charges from initial +0.7 V toward -Vsat:

vc(t) = -Vsat + [0.7 - (-Vsat)] * exp(- t / RC) approx -Vsat * [1 - exp(- t / RC)].

- When vc(t) crosses V_LT = - beta * Vsat at t = T_pulse, op-amp switches back to +Vsat.
- Equating: - beta * Vsat = - Vsat * [1 - exp(- T_pulse / RC)].

exp(- T_pulse / RC) = 1 - beta => T_pulse = R * C * ln[1 / (1 - beta)].

- Circuit returns to stable state and resets.

[illegible] acquisition (ECG/EEG) and Right-Leg Drive

- This negative feedback dynamically cancels patient common-mode displacement currents, reducing 50 Hz interference by an additional 30-40 dB, ensuring artifact-free medical recordings.
